

TLDR: Ice Cream, Beach, late nights

Dear members of the Baja extended family

Welcome to the New York edition of the McGill Baja Friends and Alumni newsletter! As another competition season passes, there are new things that we need to get you up to speed on. This is our voyage down to Hogs Hill in upstate New York.

As in previous editions of the newsletter, technical sections will be in **[red]** while general explanations will be in **[blue]**.

Prologue: Pre-Competition

As the team prepared for this competition, we set multiple hard deadlines which would allow for our car to be thoroughly tested before competition (affectionally coined “comp”). With our expected timeline, we had a rolling car deadline for April 1st to have a month and a half to tune car parameters for a better performing car. Fitting with the theme of the holiday that this day falls upon, the car was finished June 10th at 7 am with plans to leave for comp at 8 am the same day.



Preparing the car for comp

Day 0: Leaving

As the people who stayed up all night finishing the car went back home to pack, the first cars prepared to leave. It started out with Yasmine's car leaving punctually at 8 am, then Scott's car

next at 9:00 am after completing a 20-point turn to back out of a parking spot blocked by the trailer. They were quickly followed by Frank's, Patricia's, and Zac's cars and the truck.

Once the cars left school, thus began the 6-hour journey to the hotel, which led to an unexpected world tour, passing through Mexico and right by Greece. Now for a few highlights from each of the car's adventures.



Not to be confused with Mexico City

Yasmine's car stopped at Trader Joe's to get lunch before heading to the Ontario beach.

Scott's car got stuck in a rainstorm on the way down, stopped at Chipotle for lunch, went in to Trader Joe's for snacks, and picked up Ariel at a small café where he had been waiting for 4 hours. After checking, Ariel could have walked to the hotel from where he was in 3 hours.

Frank's car was the first to arrive at the hotel, having time to check in, drop off their bags, head out for dinner, and come back before most of the other cars had even arrived.

Patricia's car stopped for lunch in Brockville on the way down and visited an old train tunnel before passing by the lake shore for an afternoon stroll.



Patricia's car

Zac's car stopped at Chez Flamand's for breakfast before passing on to the border pretty early on but was taken in for secondary inspection. They were then stuck at the border for 2 hours, giving them just enough time to watch the truck pass through without any issues.

Finally, as the truck was the last to leave, they made a straight shoot down to Rochester.

Once everyone had arrived, we set up the car in the hotel parking lot with two other teams, one who had come all the way from Mexico City! The rest of the night had people split into two groups; those making sure the car would be ready for engine check the next morning, and those rushing to finish their poster for the Design Review presentations on the second day of comp.

Day 1: Engine Check and Technical Inspection

The competition is divided into static events (engine check, technical inspection, design presentations, and brake check) which ensure that your vehicle is safe to be driven while also evaluating the design of your baja, and dynamic events (acceleration, hill climb, maneuverability, suspension and traction, and endurance) where the car's performance is tested.

The main activity for the first day of the competition was passing engine check where the judges make sure that your car's top RPM does not surpass a set value. Out of over 100 teams, we were 20th in line and passed the first time! We then continued to pass technical inspection (tech) but mid-way through, a severe weather warning was issued leading to the organizers closing tech for the rest of the day. The site was then closed, and teams were told to evacuate, forcing us to leave our car on the comp site despite having plans to run the car in the parking lot that night.

To keep spirits up, we swarmed a local ice cream shop before the comp organizers announced that they were reopening the site. To prepare for brake check the following day, the brakes and electronics teams returned to the site.



The team's quick ice cream run

The rest of the team headed back out for supper, with Scott and Zac's car heading to a pizza place. Notable moments from the journey there were getting kicked out of a liquor store, an underage person left in the car while the 'adults' went to buy bourbon (is there a pattern appearing), and a driver's license left behind at a liquor store.

Once everyone returned to the hotel, the team focused its efforts on practicing for design presentation before turning in for an early night (1am) so that everyone was ready to be on site for 8 am.

Day 2: Technical Inspection continues and Brake check

Starting out, right after arriving at the comp site, the team raced to get our car to the tech tents to finish our general and final inspection which we passed on our first try, something unprecedented which hasn't happened in years!

Due to our tight schedule, the leads and captains rushed to the next tent in our fancy McGill Baja polos and black pants so we could give our design presentations (design prez).



SSB Design Prez



Captain Jemma Design Prez

Taking advantage of our good luck, we went to brake check. During our first attempt, the **car veered sides while braking and we received a warning that the seat belt pointed at a downward angle** on our driver's shoulder which could lead to penalties. We realized one of our front tires was flat which explained why the car had veered right while braking and ended up switching to a taller driver to avoid any repercussions due to the seat belt issue, leading to a successful second run.

After flying through the static events, the team got some time in at the practice track so drivers could get a feel for the car on rough terrain and also helping us realize that unfortunately, **we still have a large turning radius, meaning that the car would have trouble taking sharp corners because it needed more space to complete turns. One solution to this is having more oversteer or the ability to break traction in the rear which would allow the car to drift in tight corners, something our car also couldn't do.** With the power of foresight, we had at least designed our car with reverse, so we'd still be able to get out of these tricky situations.



Scenic picture of the baja

With all the major events passed, we were left to put the finishing touches on our car, but this led to us realizing that **because of the tight fit of our bearing in the upright, they were not pressed incompletely so we had not been able to put a retaining ring on our hubs, leading to the upright un-pressing itself from the hub. The force of the hub un-pressing itself ended up bending the brake discs on the front**, leading to another late-night repair. The inside of the upright was filed down, the bearings were dunked in an ice bath, and four people were needed to press the bearings into the uprights.



How many engineers does it take to press a bearing

Day 3: Dynamic Events

Thus began one of the most exciting days of comp, finally putting our car to the test, but first, we had to put our tent to the test...which, well you can see what happened.



The tent snapping in half

After that mishap we finally moved on to the dynamic events of acceleration, hill, climb, maneuverability, and Suspension & Traction (S&T). To give a quick recap; Acceleration: due to being unable to tune the CVT, our top speed wasn't the best, Hill Climb: with a functioning 4WD, we crushed this event, Maneuverability: we completed the track with the help of reverse, S&T: nothing broke (that we know of) and we almost finished the track!

After an eventful day, there was still a little work left to be done on the car because the electronics team continued to work on the DAQ system and the primary clutch wasn't coming off the engine shaft... After a ton of stress and help from 3 other teams we finally got it off. Because of this, our reservation for Olive Garden (originally for 8 pm) got pushed to 9, then 9:30, but luckily everyone arrived and we had team dinner where we celebrated Jemma's (at least) 12th and final birthday with the team.



Team dinner

A big shout out to the Olive Garden staff there that night because they kept the restaurant open and waited for us to finish eating before closing and didn't mind us having the baja in the truck bed.



Strapped and loaded

With things still left to be done on the car, the CVT and electronics teams stayed up working in the parking lot while the rest of the team drove to Ontario Beach for a late-night dip.



Making quite the splash

Day 4: Endurance

On our final day of comp, the team divided into two, those part of Carnage crew, a group responsible for helping cars when they get stuck anywhere on the track, and pit crew, those that stay back at the paddock in case the car needs to refuel or undergo repairs.

Endurance lasts 4 hours, but unfortunately, the car wasn't running that whole time due to a throttle cable snapping, running out of fuel on the track, a visor flying off one of the driver's helmets, and two half shafts sheering.

We began in 52nd, then climbed up approximately 30 positions before our first failure which dropped us down to around 40th. This pattern repeated itself where failures would occur after the baja had managed to move up to a spot in the 20s and we would drop back down. To put things into perspective, our fastest lap time was 3:40, making us the 17th fastest car in terms of best lap time. Despite all our issues, McGill Baja placed 45th out of 102 cars in the endurance race.

Epilogue: Returning to Montreal

After a week full of action, some members of the team headed back to their jobs or homes in the states, others continued to visit Rochester for another few days, and the rest returned to

Montreal to unpack the trailer, clean up the shop, and review the aftermath of the competition. Looking back, with the car finished right in the nick of time, we're thinking of calling her Nicky.



MB27 Group Picture



Scott's car took the scenic route, passing right by Madrid

It is with heavy hearts that we must say goodbye to our senior captain, Jemma Pooley. After years of dedication to McGill Baja, she will be moving on and we wish her luck in her future ventures as Design Team Director which means she hasn't seen the last of us and neither have you.

Until next time,

Rhea Duguay



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